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# **DATABASE**

**PREPARATION FOR THE FINAL DATABASE PRACTICAL EXAM**

**Author:**

Dr. Ir. Agus Eko Minarno, S.Kom., M.Kom., IPM.

Naufal Arkaan

Ismail Dwi Muh. Anugerah

**INFORMATICS LABORATORY**

**UNIVERSITY of MUHAMMADIYAH MALANG**

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## INTRODUCTION

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### OBJECTIVES

1. Understanding how to use **Oracle APEX** as a medium for conducting Final Practicum Exams.
2. Knowing the provisions, rules, and guidelines for conducting the **Final Database Practicum Exam**.
3. Prepare yourself technically and conceptually before taking the Final Practicum Exam.

### MODULE TARGETS

1. Able to operate **Oracle APEX** independently according to the steps described in the module.
2. Be able to prepare yourself and understand the flow of the **Database Practicum Final Exam** in accordance with applicable regulations.

### PREPARATION

1. Muslim students may recite the following prayer before starting the exam

رَضِيتُ بِاللّهِ رَبًّا وَبِالْإِسْلَامِ دِينًا وَبِمُحَمَّدٍ نَبِيًّا وَرَسُولًا. رَبِّ زِدْنِي عِلْمًا وَارْزُقْنِي فَهْمًا

2. Laptop or personal computer
3. Stable internet connection
4. Oracle APEX Workspace accounts that have been activated in accordance with the naming conventions

### KEYWORDS

Oracle Apex, SQL, Query



## FINAL EXAM REQUIREMENTS FOR DATABASE

### General Provisions

1. The Final Practicum Exam (UAP) is the final evaluation of the **Database** Course practicum.
2. The UAP is conducted **offline** in the Computer Laboratory according to each practical class schedule.
3. The exam is conducted using the **Oracle APEX** web-based platform as a medium for answering questions.

### Technical Provisions for Implementation

1. Students work on UAP using the **laboratory computers** provided.
2. Students are **required to bring personal backup devices** (laptops or other devices) in anticipation of technical problems, as well as to **set up personal hotspots** in case of internet or campus WiFi network disruptions.
3. The Oracle APEX workspace must be created before the exam day and use the naming format according to the guidelines in the How to Use Oracle APEX section, for the naming format **(last 3 digits of student ID\_DATABASEEXAM\_2026)**  
**(example: 123\_DATABASEEXAM\_2026)**.
4. The exam is a live query, with the material covered in **modules 4 to 6**.
5. Technical issues due to device, account, or internet connection problems are **not grounds for extending the exam time**.
6. The UAP exam lasts 50 minutes and begins simultaneously according to the supervisor's instructions.

### Academic Integrity Policy

1. Students are **prohibited from bringing or opening notes**, either in physical or digital form, during the examination.
2. Students are **prohibited from communicating**, collaborating, or sharing information with other students in any form.
3. The Oracle APEX workspace used **must be empty** before the exam begins.
4. Violations of these provisions will result in a reduction in marks or even cancellation of the exam, depending on the severity of the violation.

### Punctuality and Discipline

1. Students **must arrive on time** according to the predetermined practicum schedule.
2. Students who **leave the examination room without the supervisor's permission are considered to have withdrawn from the examination**.



### Closing

1. All students **must comply with all UAP regulations** that have been established.
2. These terms are **binding and non-negotiable**.
3. Violations of these provisions will be subject to **sanctions in accordance with applicable practicum policies**.

### EXAM GRADING RUBRIC

| Assessment Aspects               | Points            |
|----------------------------------|-------------------|
| <b>Final Exam (LIVE QUERY)</b>   | <b>Total 100%</b> |
| DATA MANIPULATION LANGUAGE (DML) | 10%               |
| FILTER                           | 15%               |
| AGGREGATE FUNCTION               | 20%               |
| JOIN                             | 20%               |
| SUB QUERY                        | 25%               |
| UNDERSTANDING MATERIAL           | 10%               |



## EXAM CONTENT OVERVIEW

### Examined Material

The Final Exam covers **Modules 4 to 6**. Exam questions are presented in **table format**, so that the answers collected by students are only the **results of SQL queries** executed according to the instructions in the questions.

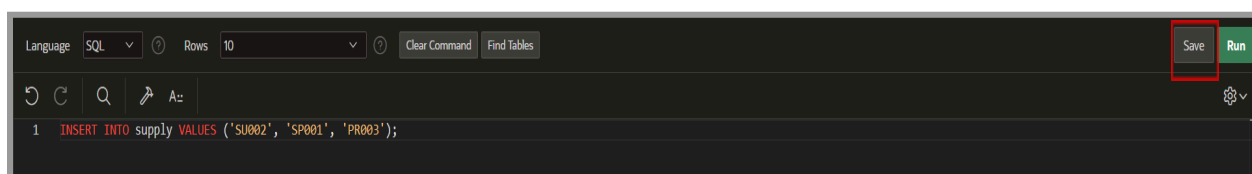
### Final Practical Exam Material Coverage

- The following are the sub-topics that will be tested during the UAP:
  - **DATA MANIPULATION LANGUAGE (DML)**
  - **FILTER**
  - **AGGREGATE FUNCTION**
  - **JOIN**
  - **SUB QUERY**

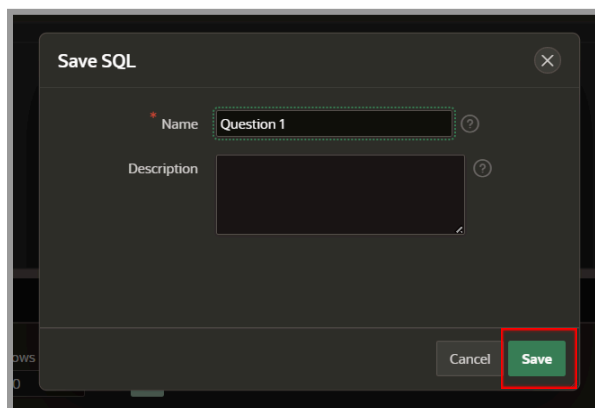
### How to Save a Query in Oracle Apex

During the Final Database Practicum Exam, after the participants have finished typing and running the queries for each question in Oracle APEX, they are expected to save the queries that they consider to be correct and final, in order to facilitate the assistants in assessing the query results that have been completed.

- Click the save button



- Enter a name and then click the save button



3. If successful, the query you saved will appear in the **Saved SQL** tab.

Results

Explain

Describe

Saved SQL

History

Owner ⓘ

- All Users -

Find ⓘ

Rows

10

Go

Delete Checked

| <input type="checkbox"/>            | Owner                     | Name       | Description | SQL                                                    | Updated By                | Updated      |
|-------------------------------------|---------------------------|------------|-------------|--------------------------------------------------------|---------------------------|--------------|
| <input checked="" type="checkbox"/> | NAUFALARKAAN875@GMAIL.COM | Question 1 | -           | INSERT INTO supply VALUES ('SU002', 'SP007', 'PR003'); | NAUFALARKAAN875@GMAIL.COM | 1seconds ago |

## Practice Questions

Based on the results in the table below, write the appropriate SQL query. Ensure that the output is identical to the table shown.

### 1. QUESTION 1 – DML (INSERT DATA)

**Table: student**

| student_id (VARCHAR) | student_name (VARCHAR) |
|----------------------|------------------------|
| S001                 | Andi                   |
| S002                 | Budi                   |

**Table: lecturer**

| lecturer_id (VARCHAR) | lecturer_name (VARCHAR) |
|-----------------------|-------------------------|
| L001                  | Dr. Raka                |
| L002                  | Dr. Sinta               |

**Table: project**

| project_id (VARCHAR) | project_title(VARCHAR) | field (VARCHAR) |
|----------------------|------------------------|-----------------|
| P001                 | AI Detection System    | AI              |
| P002                 | Database Optimization  | Database        |



**Table: supervision**

| supervision_id<br>(VARCHAR) | student_id<br>(VARCHAR) | lecturer_id<br>(VARCHAR) | project_id<br>(VARCHAR) |
|-----------------------------|-------------------------|--------------------------|-------------------------|
| SV01                        | S001                    | L001                     | P001                    |
| SV02                        | S002                    | L002                     | P002                    |

**Table: evaluation**

| evaluation_id (VARCHAR) | supervision_id (VARCHAR) | score (NUMBER) |
|-------------------------|--------------------------|----------------|
| E001                    | SV01                     | 85             |
| E002                    | SV02                     | 90             |

## 2. QUESTION 2 – FILTER (Simple Filter)

| student_name | project_title       | score |
|--------------|---------------------|-------|
| Andi         | AI Detection System | 85    |

## 3. QUESTION 3 – AGGREGATE FUNCTION (Combining)

| field    | average_score |
|----------|---------------|
| AI       | 85            |
| Database | 90            |

## 4. QUESTION 4 – JOIN (INNER JOIN)

| lecturer_name | student_name | project_title         |
|---------------|--------------|-----------------------|
| Dr. Raka      | Andi         | AI Detection System   |
| Dr. Sinta     | Budi         | Database Optimization |



## 5. QUESTION 5 – MULTI-ROW SUBQUERY

| student_name |
|--------------|
| Andi         |
| Budi         |

### Answer 1 – DML (INSERT DATA)

```
CREATE TABLE student (
  student_id VARCHAR2(5) PRIMARY KEY,
  student_name VARCHAR2(50)
);

CREATE TABLE lecturer (
  lecturer_id VARCHAR2(5) PRIMARY KEY,
  lecturer_name VARCHAR2(50)
);

CREATE TABLE project (
  project_id VARCHAR2(5) PRIMARY KEY,
  project_title VARCHAR2(100),
  field VARCHAR2(30)
);

CREATE TABLE supervision (
  supervision_id VARCHAR2(5) PRIMARY KEY,
  student_id VARCHAR2(5),
  lecturer_id VARCHAR2(5),
  project_id VARCHAR2(5),
  CONSTRAINT fk_student
    FOREIGN KEY (student_id) REFERENCES student(student_id),
  CONSTRAINT fk_lecturer
    FOREIGN KEY (lecturer_id) REFERENCES lecturer(lecturer_id),
  CONSTRAINT fk_project
    FOREIGN KEY (project_id) REFERENCES project(project_id)
);

CREATE TABLE evaluation (
```



```

evaluation_id VARCHAR2(5) PRIMARY KEY,
supervision_id VARCHAR2(5),
score NUMBER,
CONSTRAINT fk_supervision
    FOREIGN KEY (supervision_id) REFERENCES supervision(supervision_id)
);

INSERT INTO student VALUES ('S001', 'Andi');
INSERT INTO student VALUES ('S002', 'Budi');

INSERT INTO lecturer VALUES ('L001', 'Dr. Raka');
INSERT INTO lecturer VALUES ('L002', 'Dr. Sinta');

INSERT INTO project VALUES ('P001', 'AI Detection System', 'AI');
INSERT INTO project VALUES ('P002', 'Database Optimization', 'Database');

INSERT INTO supervision VALUES ('SV01', 'S001', 'L001', 'P001');
INSERT INTO supervision VALUES ('SV02', 'S002', 'L002', 'P002');

INSERT INTO evaluation VALUES ('E001', 'SV01', 85);
INSERT INTO evaluation VALUES ('E002', 'SV02', 90);

```

## Answer 2 – SIMPLE FILTER

```

SELECT s.student_name,
       p.project_title,
       e.score
FROM evaluation e
JOIN supervision sv ON e.supervision_id = sv.supervision_id
JOIN student s ON sv.student_id = s.student_id
JOIN project p ON sv.project_id = p.project_id
WHERE p.field = 'AI'
AND e.score >= 80;

```

## Answer 3 – COMBINING AGGREGATE FUNCTION

```

SELECT p.field,
       AVG(e.score) AS average_score
FROM evaluation e

```



```
JOIN supervision sv ON e.supervision_id = sv.supervision_id
JOIN project p ON sv.project_id = p.project_id
GROUP BY p.field;
```

#### Answer 4 – JOIN (INNER JOIN)

```
SELECT l.lecturer_name,
       s.student_name,
       p.project_title
FROM supervision sv
INNER JOIN lecturer l ON sv.lecturer_id = l.lecturer_id
INNER JOIN student s ON sv.student_id = s.student_id
INNER JOIN project p ON sv.project_id = p.project_id;
```

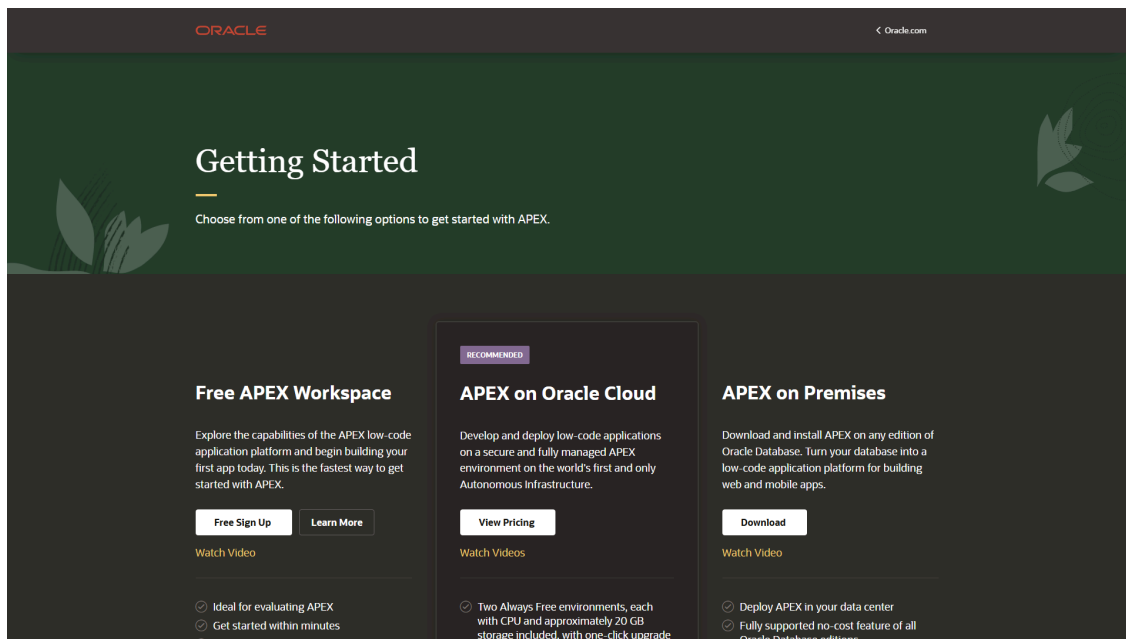
#### Answer 5 – MULTI-ROW SUBQUERY

```
SELECT student_name
FROM student
WHERE student_id IN (
    SELECT sv.student_id
    FROM supervision sv
    JOIN evaluation e ON sv.supervision_id = e.supervision_id
    WHERE e.score >= (
        SELECT AVG(score)
        FROM evaluation
    )
);
```

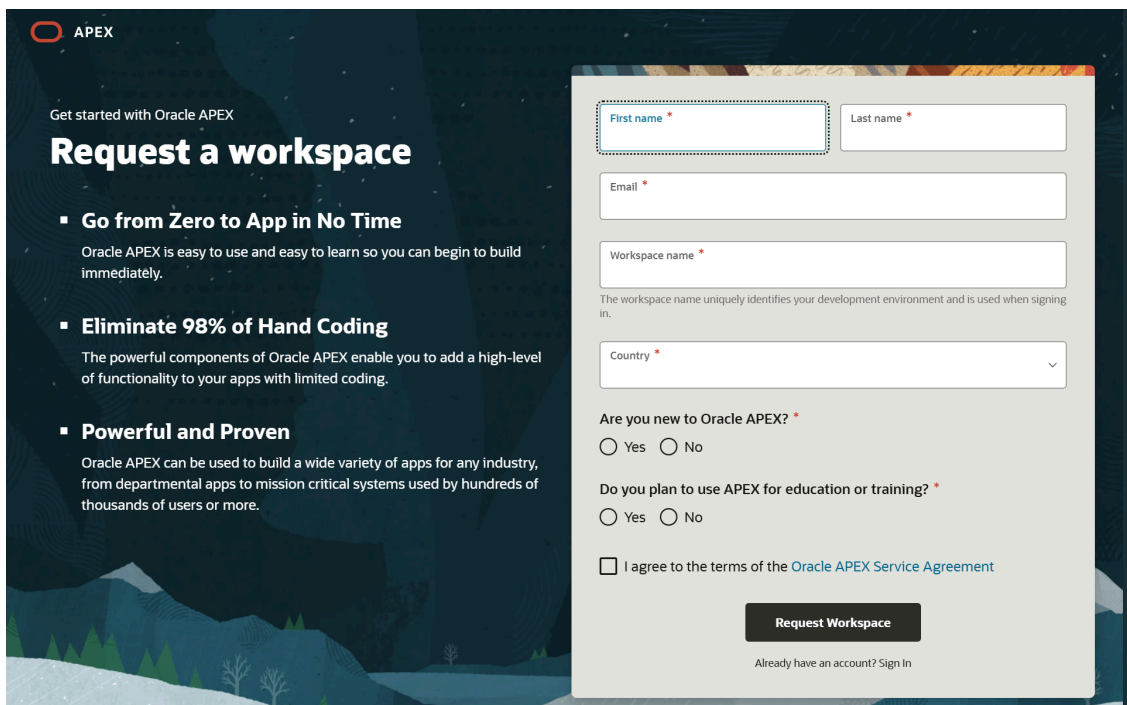


## HOW TO USE ORACLE APEX

1. Go to <https://apex.oracle.com/en/learn/getting-started/>



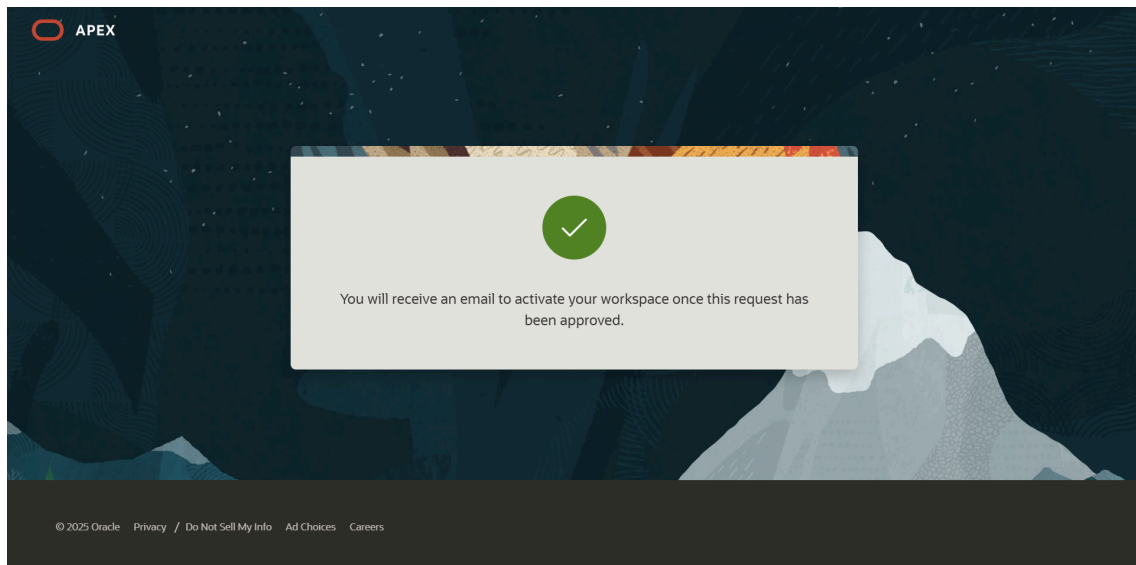
2. Click "Free Sign Up".



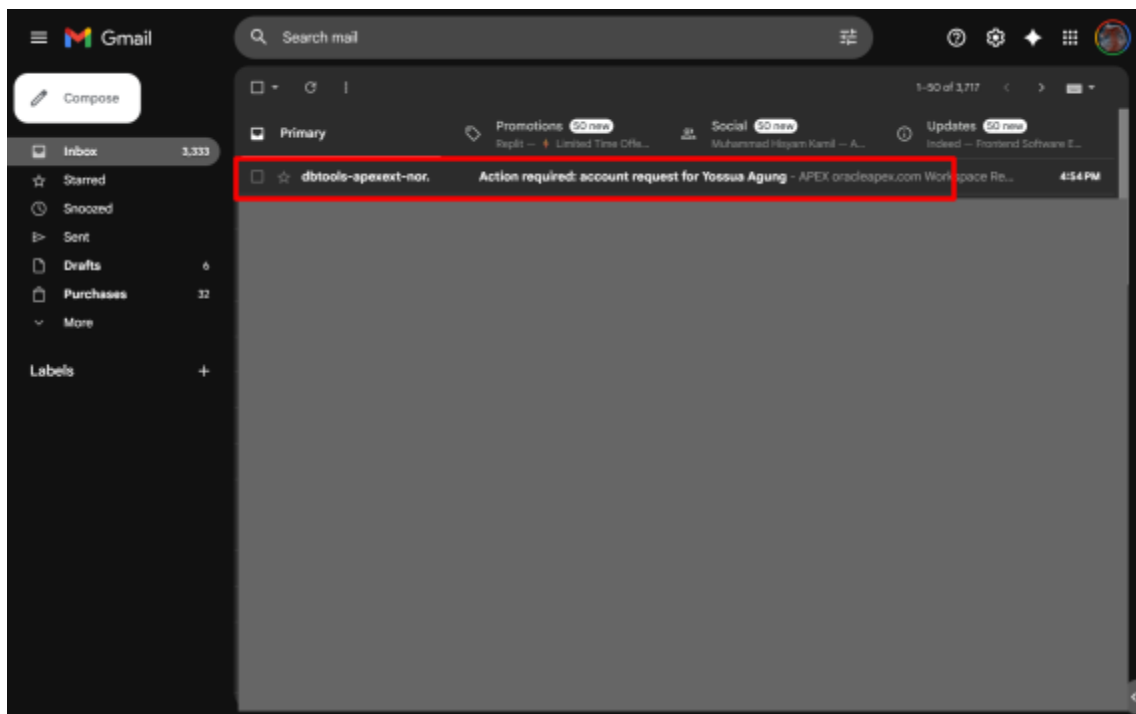
3. Fill in your personal data. Use the workspace name format: [Last 3 Digits of Student ID\_DATABASEEXAM\_2026]. Example **123\_DATABASEEXAM\_2026**.



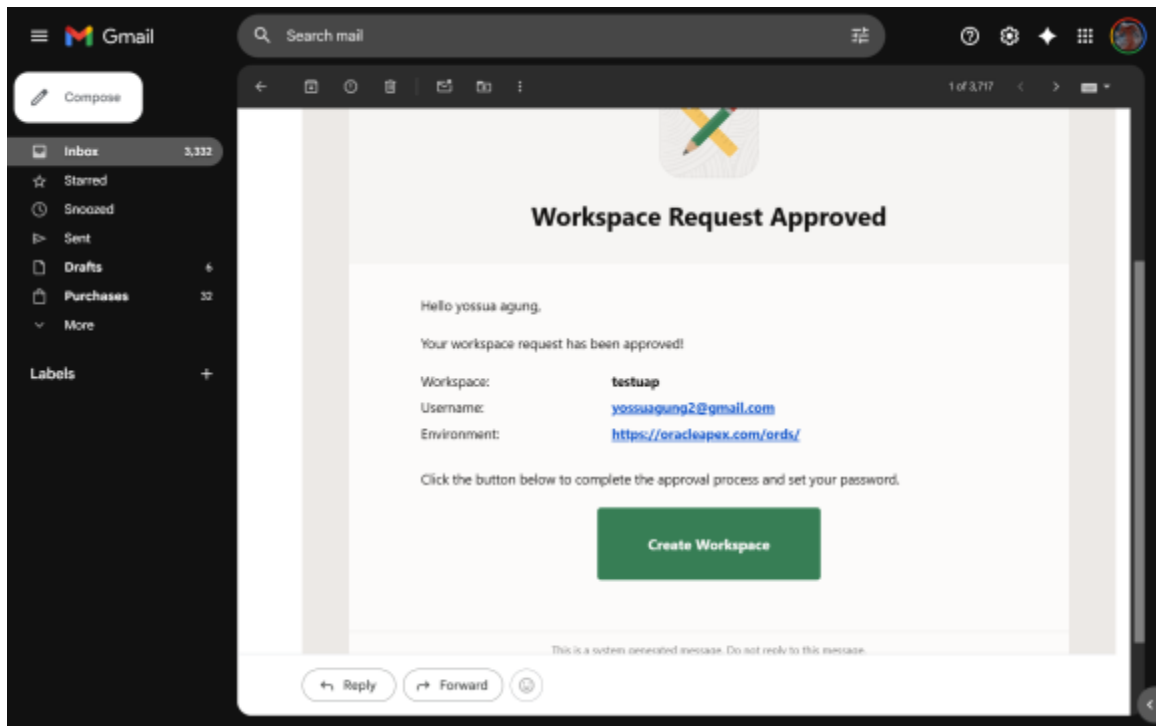
- You will receive an email to activate your workspace.



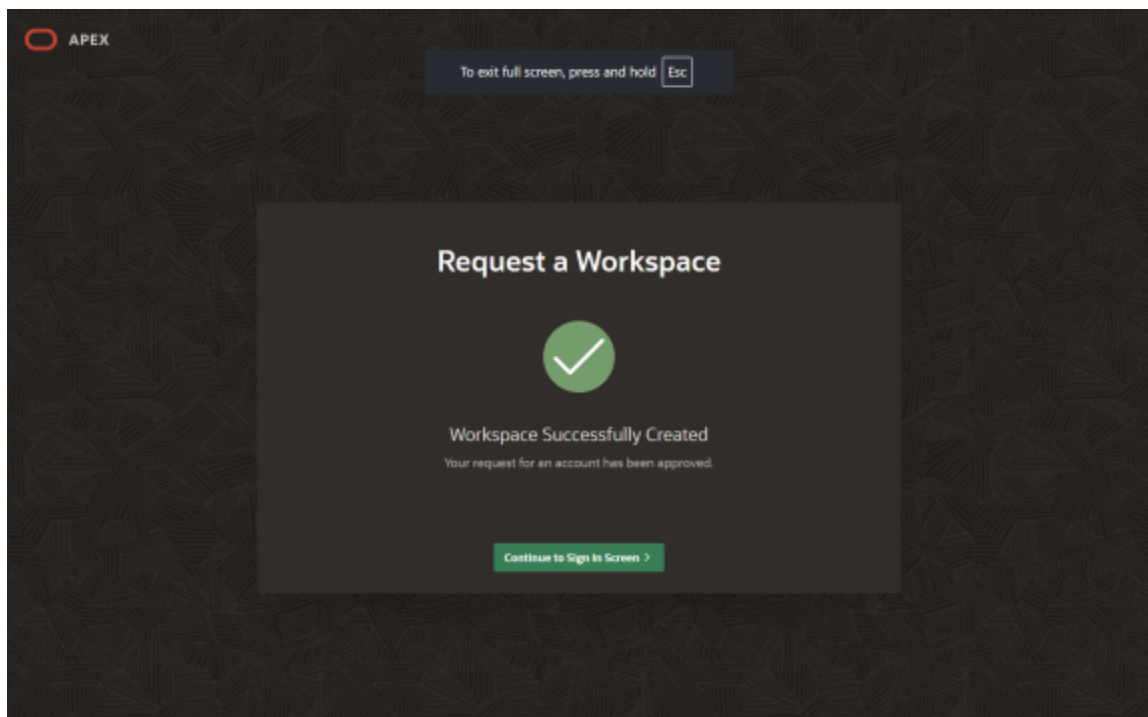
- Open your email, then look for an email with a name like the following.



- Open the email, then scroll down until you find the text "Create Workspace".

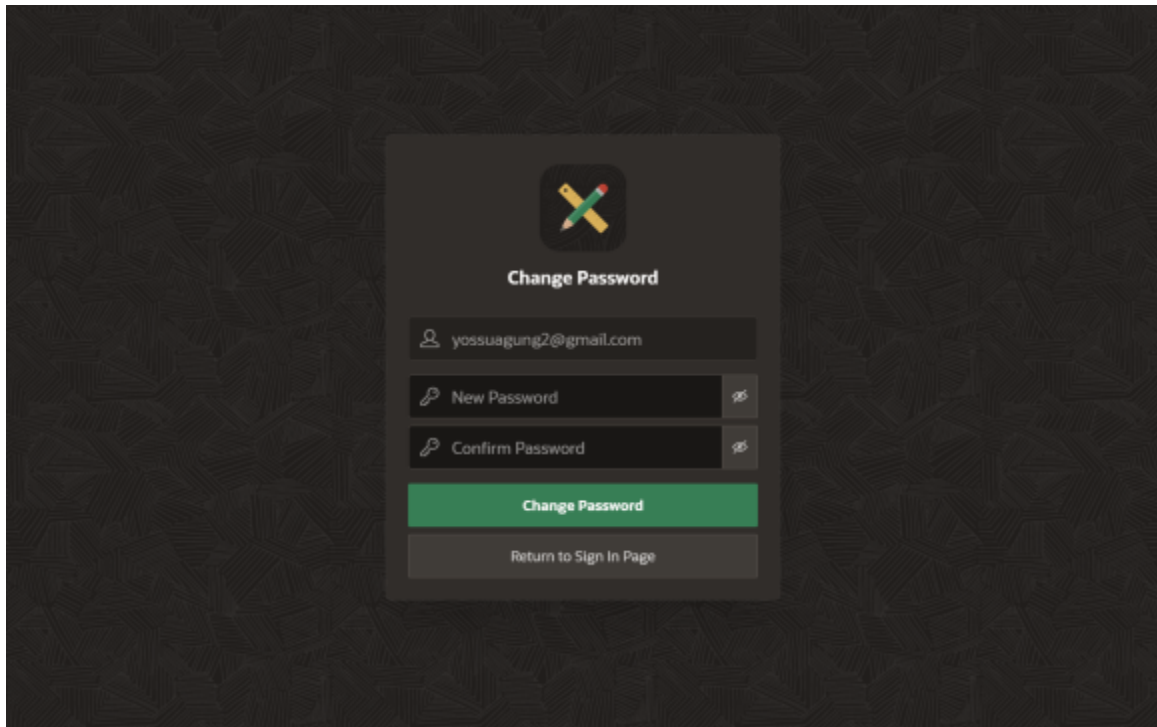


- Click and you will be directed to a new tab, and there will be a display like the following.

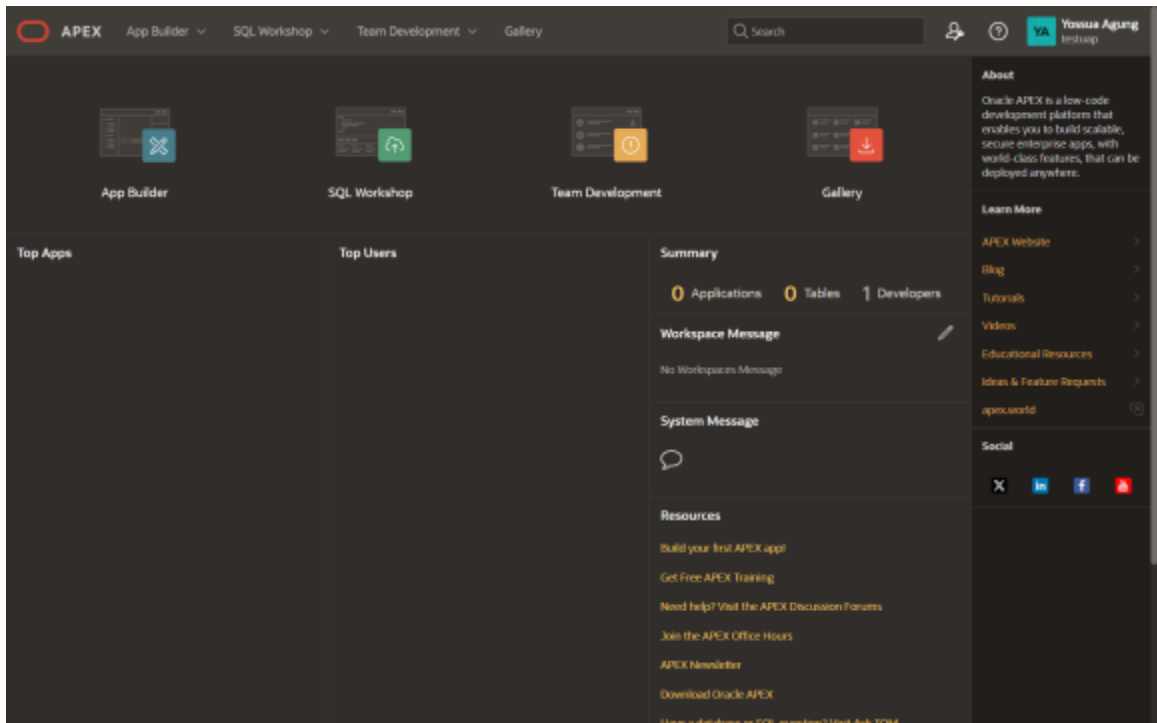


- Click "Continue to Sign In Screen" and you will be directed to fill in a new password, fill it in with your complete Student ID number (NIM).





9. If it's done, you will be directed to the Oracle APEX initial screen.

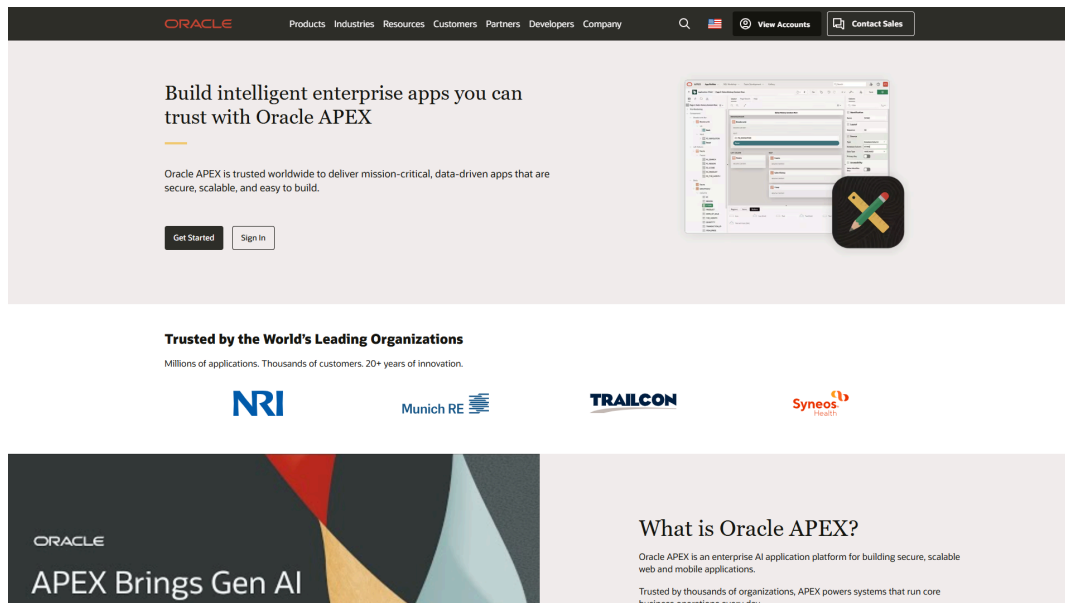


It should be noted that later for login you must use the Email you used when registering for Oracle Apex, and your complete Student ID number (NIM) for the password, so please, if you do not memorize each of your Student ID numbers, please write it down on paper or on your phone, and it can be typed when in the lab.

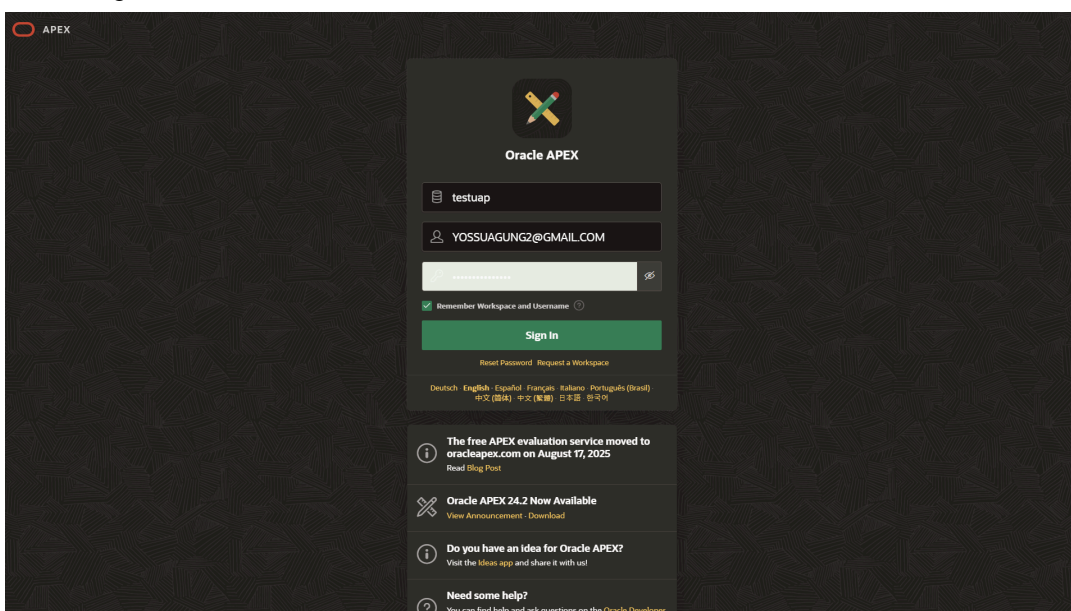


## How to log in to Oracle APEX.

1. Go to [www.oracleapex.com](https://www.oracleapex.com).



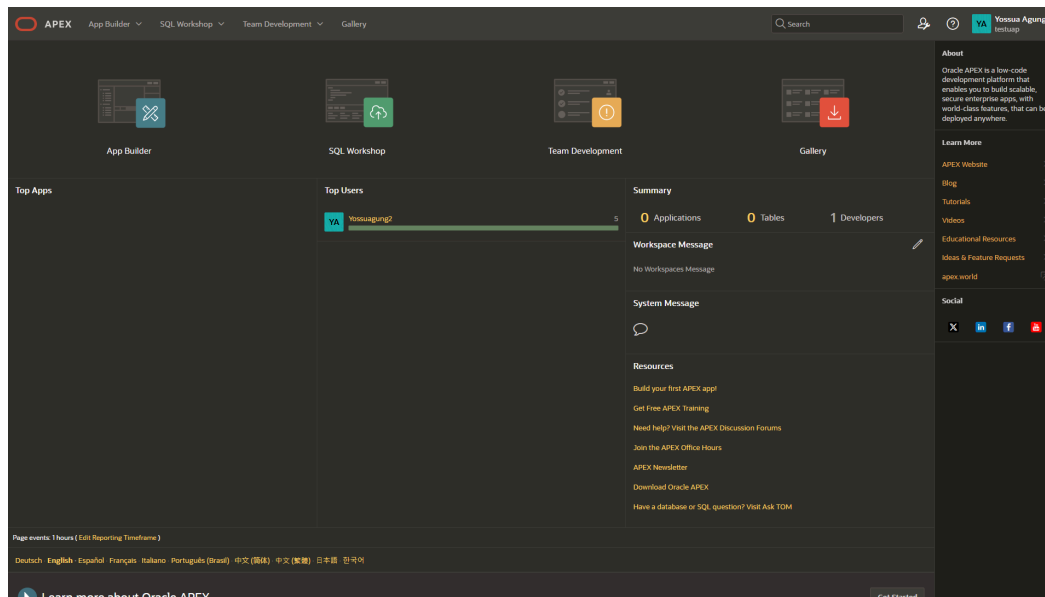
2. Click Sign in.



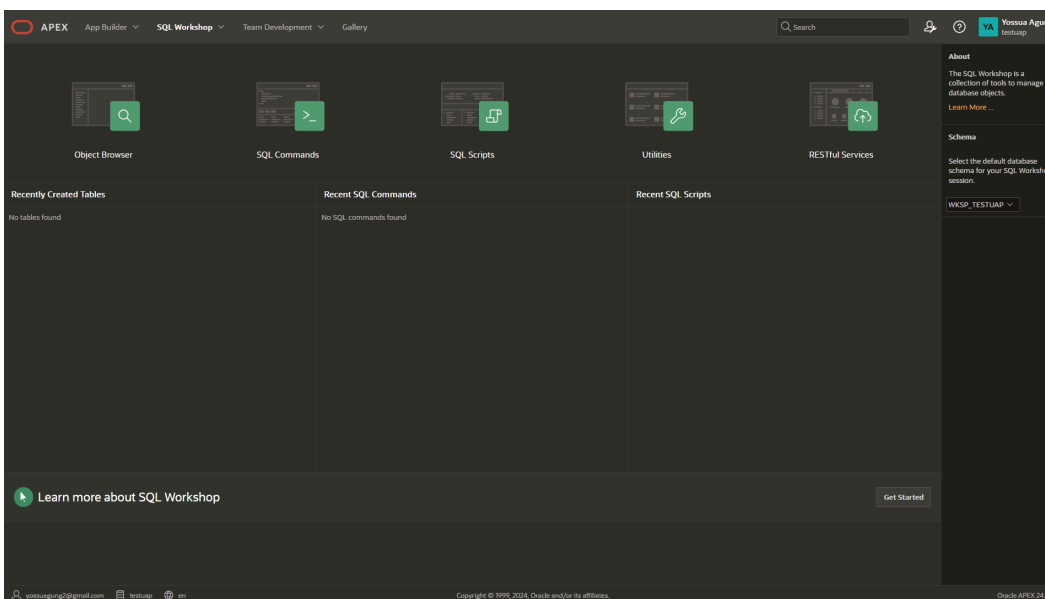
3. Enter the workspace name, the email used, and the password that was created during the workspace request. Then click sign in.



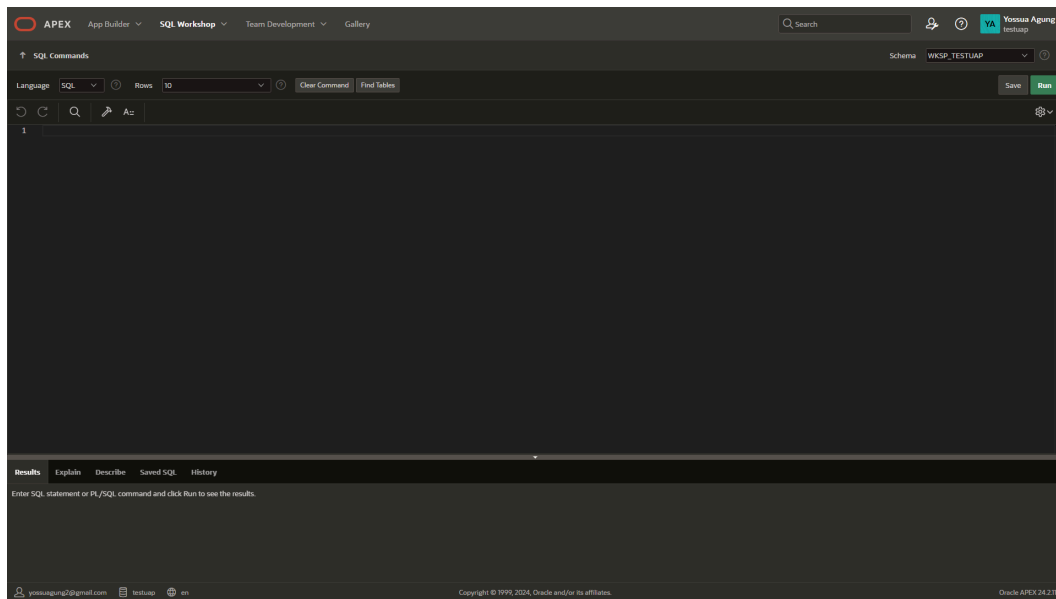
- If it's done, you will be directed to the Oracle APEX initial screen.



- To use sql commands you can click the sql workshop.



6. Then click sql commands.



You can now use the Web-based Oracle Apex.

